

D2AN: Delay-Doppler Attention Network for Channel Estimation in OFDM Systems

Baichuan Zhao^{ID}, *Member, IEEE*, Qi Sun^{ID}, *Member, IEEE*, Zhi Wang, Nan Li^{ID}, *Member, IEEE*,
and Chih-Lin I^{ID}, *Life Fellow, IEEE*

Abstract—Accurate channel estimation is essential for fully exploiting the advantages of orthogonal frequency division multiplexing (OFDM) systems. Compared with conventional estimators, deep neural networks can achieve higher estimation accuracy. Nevertheless, most existing deep learning-based approaches are purely data-driven, which limits their ability to effectively capture the inherent channel correlations. As a result, their performance degrades in dynamic environments and their interpretability remains limited. To address these challenges, this letter proposes a novel architecture, referred to as the Delay-Doppler Attention Network (D2AN). In particular, we design an attention mechanism where the attention coefficients are generated from basis functions sampled in the delay and Doppler domains. By incorporating the structure of channel correlation functions into the attention layers, D2AN explicitly models the underlying spectral and temporal dependencies of wireless channels. Simulation results demonstrate that the proposed D2AN significantly outperforms state-of-the-art deep learning-based counterparts as well as conventional channel estimators. Moreover, D2AN exhibits strong robustness across diverse signal-to-noise ratios, user mobilities, and delay spreads, while maintaining low computational complexity.

Index Terms—Channel estimation, OFDM, deep learning, delay-Doppler domain, attention mechanism.

I. INTRODUCTION

CHANNEL estimation is essential in orthogonal frequency division multiplexing (OFDM) systems for reliable data recovery and high throughput [1]. Traditionally, this is achieved via least squares (LS) estimation [2] or linear minimum mean square error (LMMSE) estimation [3]. However, LS fails to exploit channel statistics and suffers from inferior performance, whereas LMMSE achieves higher accuracy at the expense of requiring prior channel statistics that are hard to obtain in practice.

Deep neural networks (DNNs) have been employed to overcome these limitations. Early works mainly rely on convolutional neural networks (CNNs) and recurrent neural networks (RNNs). For example, ChannelNet treats the channel estimation problem as a super-resolution task, and applies two cascaded super-resolution and denoising CNNs to refine the coarse interpolated estimate [4]. To reduce the computational burden, ReEsNet performs upsampling at the final stage of

the network [5]. To capture long-range dependencies, RNNs were cascaded with CNNs in [6] and [7] to capture temporal and spectral channel correlations, respectively, while attention mechanisms were integrated with CNNs in [8] and [9]. More recently, Transformers have been applied to channel estimation. In [10] and [11], the channel matrix was partitioned into several sub-matrices, which were treated as distinct tokens. To enhance robustness, environmental characteristics such as maximum Doppler shift and delay spread were treated as auxiliary inputs [10], [11].

Despite the improved performance, existing deep learning methods struggle to capture complex channel correlations effectively. Specifically, they operate in a purely data-driven manner, i.e., treating the communication system as a closed box without incorporating prior knowledge from wireless communications. This nature leads to high training costs and poor interpretability and robustness [12]. Although some recent works incorporated channel parameters as extra inputs [10], [11], such integration at the feature level is not enough to handle diverse channel conditions. As a result, their performance often degrades when the environment changes significantly.

In this letter, we propose a model-driven DNN, namely Delay-Doppler Attention Network (D2AN), which bridges statistical signal processing and deep learning for channel estimation. Inspired by the channel autocorrelation function, we design a novel attention mechanism that embeds domain knowledge. This design enables D2AN to effectively capture channel correlations and improve estimation accuracy. The main contributions of this work are summarized as follows:

- We propose a novel attention mechanism by incorporating the structure of channel correlation functions. The attention coefficients are generated from basis functions sampled in the delay and Doppler domains. The mechanism can effectively capture intrinsic temporal and spectral correlations with enhanced interpretability.
- Building upon the proposed attention mechanism, we develop the D2AN. Extensive simulation results demonstrate that the proposed model achieves more accurate channel estimation than state-of-the-art benchmarks.
- We demonstrate the robustness of the proposed model across a wide range of channel conditions and interpret its superiority through visualization of the attention matrices.

Notation: $\odot$ denotes Hadamard product. $\|\cdot\|_F$ denotes the Frobenius matrix norm. $|\cdot|$ denotes absolute value of scalars. $(\cdot)^T$ denotes transpose of matrices. $\mathbf{I}$ denotes identity matrix. $\text{vec}(\cdot)$ denotes the operation of concatenating each column of a matrix into a vector.

Received 3 April 2026; revised 9 May 2026; accepted 14 May 2026. Date of publication 20 May 2026; date of current version 2 June 2026. This work was supported by the National Science and Technology Major Project of China under Project GXB-2-2025-5. The associate editor coordinating the review of this article and approving it for publication was W. Liu. (*Corresponding author: Qi Sun.*)

The authors are with China Mobile Research Institute, Beijing 100034, China (e-mail: zhaobaichuan@chinamobile.com; sunqiyjy@chinamobile.com; wangzhi@chinamobile.com; linan@chinamobile.com; icl@chinamobile.com). Digital Object Identifier 10.1109/LWC.2026.3695069

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a single-input single-output (SISO) OFDM communication system characterized by N_f subcarriers and N_t OFDM symbols as in [4], [5], [6], [7], [10], and [11], to facilitate a fair comparison and for clarity. The received signal is

$$\mathbf{Y} = \mathbf{H} \odot \mathbf{X} + \mathbf{Z} \quad (1)$$

where $\mathbf{Y}, \mathbf{H}, \mathbf{X}, \mathbf{Z} \in \mathbb{C}^{N_f \times N_t}$ represent the received symbol, channel matrix, transmitted symbol and noise, respectively, $z_{k,n} \sim \mathcal{CN}(0, \sigma^2)$ and σ^2 is the noise power.

To facilitate channel estimation, pilot symbols are inserted into the time-frequency grid according to a specific pattern. Denote $\mathbf{H}_p \in \mathbb{C}^{N_{f,p} \times N_{t,p}}$ as the submatrix of $\mathbf{H}$ containing the channel coefficients at the pilot positions. The observation at these locations can be expressed in matrix form as:

$$\mathbf{Y}_p = \mathbf{H}_p \odot \mathbf{X}_p + \mathbf{Z}_p, \quad (2)$$

where $\mathbf{X}_p, \mathbf{Y}_p$ and $\mathbf{Z}_p$ are the corresponding submatrices of $\mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$, respectively. The goal is to estimate the full channel matrix $\mathbf{H}$ based on the sparse observation matrix $\mathbf{Y}_p$.

B. Problem Formulation

The channel estimation problem can be formulated as minimizing the mean square error (MSE) between the true channel and its estimate, i.e.,

$$\min_{\hat{\mathbf{H}}} \mathbb{E} \left[\|\mathbf{H} - \hat{\mathbf{H}}\|_F^2 \mid \mathbf{Y}_p \right], \quad (3)$$

where the expectation is taken with respect to the conditional distribution of $\mathbf{H}$ given $\mathbf{Y}_p$, and $\hat{\mathbf{H}} \in \mathbb{C}^{N_f \times N_t}$ denotes the estimated channel matrix.

Solving (3) requires the conditional distribution of $\mathbf{H}$, which is generally unknown and difficult to characterize in practice.

C. Conventional Methods

For notational convenience, we rewrite equation (2) as

$$\mathbf{y}_p = \mathbf{X}_p \mathbf{h}_p + \mathbf{z}_p, \quad (4)$$

where $\mathbf{X}_p = \text{diag}(\text{vec}(\mathbf{X}_p))$, $\mathbf{y}_p = \text{vec}(\mathbf{Y}_p)$, $\mathbf{h}_p = \text{vec}(\mathbf{H}_p)$ and $\mathbf{z}_p = \text{vec}(\mathbf{Z}_p)$.

The LS estimator is given by $\hat{\mathbf{h}}_p^{\text{LS}} = \arg \min_{\mathbf{h}_p} \|\mathbf{y}_p - \mathbf{X}_p \mathbf{h}_p\|_F^2 = \mathbf{X}_p^{-1} \mathbf{y}_p$, and the full channel estimate $\hat{\mathbf{H}}^{\text{LS}}$ is subsequently obtained by applying interpolation methods to $\hat{\mathbf{h}}_p^{\text{LS}}$, such as bilinear interpolation or bicubic interpolation.

The LMMSE estimator is the optimal linear estimator under the MSE criterion, which is $\hat{\mathbf{h}}_{\text{LMMSE}} = \mathbf{R}_{\mathbf{h}\mathbf{h}_p} (\mathbf{R}_{\mathbf{h}_p\mathbf{h}_p} + \sigma^2 \mathbf{I})^{-1} \hat{\mathbf{h}}_p^{\text{LS}}$, where $\mathbf{R}_{\mathbf{h}\mathbf{h}_p} = \mathbb{E}[\mathbf{h}\mathbf{h}_p^H]$ and $\mathbf{R}_{\mathbf{h}_p\mathbf{h}_p} = \mathbb{E}[\mathbf{h}_p\mathbf{h}_p^H]$ represent the cross-correlation and auto-correlation matrices, respectively.

III. PROPOSED DELAY-DOPPLER ATTENTION NETWORK FOR CHANNEL ESTIMATION

Channel estimation from sparse pilots relies on leveraging the inherent correlations of wireless channels in both time and frequency domains. In this section, we propose a novel attention mechanism derived explicitly from the channel autocorrelation function. Motivated by the fact that channel correlation functions can be represented as weighted sums

of complex exponential bases, we generate attention coefficients by feeding the corresponding basis representations into fully-connected neural networks (FNNs), thereby implicitly learning the underlying channel statistics. With this mechanism, the attention weights are guided by the structure of channel correlations rather than solely learned from data, leading to more efficient, robust and interpretable learning.

A. Channel Correlation Function

To motivate the proposed method, we first revisit the channel correlation functions.

1) *Signal Model*: The channel response of a time-varying multipath channel can be modeled as

$$H(t, f) = \sum_{l=1}^L \alpha_l e^{j2\pi\nu_l t} e^{-j2\pi f \tau_l}, \quad (5)$$

where $\alpha_l, \tau_l, \nu_l \triangleq \frac{v \cos(\theta_l)}{\lambda}$ and θ_l represent the complex channel gain, propagation delay, Doppler shift and AoA of the l -th path, respectively, v is the relative velocity between the transmitter and receiver, and λ is the carrier wavelength.

2) *Temporal Correlation*: Under the standard wide-sense stationary uncorrelated scattering assumption, the temporal autocorrelation function is

$$\begin{aligned} R_t(\Delta t) &= \mathbb{E}[H(t, f)H^*(t + \Delta t, f)] \\ &= \sum_{l=1}^L \mathbb{E}[|\alpha_l|^2 e^{-j2\pi\nu_l \Delta t}] \\ &= \int_{-\nu_{\max}}^{\nu_{\max}} S(\nu) e^{-j2\pi\nu \Delta t} d\nu, \end{aligned} \quad (6)$$

where $\nu_{\max} \triangleq \frac{v}{\lambda}$, and $S(\nu)$ is the Doppler power spectrum describing the power distribution over Doppler frequencies.

3) *Spectral Correlation*: Similarly, the spectral autocorrelation function is

$$\begin{aligned} R_f(\Delta f) &= \mathbb{E}[H(t, f)H^*(t, f + \Delta f)] \\ &= \sum_{l=1}^L \mathbb{E}[|\alpha_l|^2 e^{j2\pi\Delta f \tau_l}] \\ &= \int_0^{\tau_{\max}} P(\tau) e^{j2\pi\Delta f \tau} d\tau, \end{aligned} \quad (7)$$

where $P(\tau)$ is the power delay profile and $\tau_{\max}$ denotes the maximum propagation delay spread of the channel.

B. Proposed Neural Network Architecture

Motivated by the structure of channel correlations (6) and (7), we design a neural network architecture, i.e., D2AN, that explicitly incorporates such priors into the attention mechanism for channel estimation. The goal is to guide feature interaction using meaningful correlation patterns, rather than relying solely on data-driven learning.

Similar to [5], the proposed D2AN performs most feature extraction on coarse channel representations, while upsampling is applied only at the final stage to recover the full-resolution channel. This design significantly reduces the computational cost associated with high-dimensional feature processing. The extraction is realized through a stack of Delay-Doppler attention layers, wherein the attention coefficients are generated under the guidance of both the temporal and spectral channel correlation structures. The proposed architecture is illustrated in Fig. 1 (a).

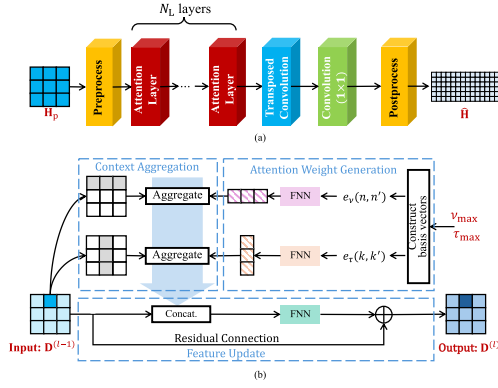


Fig. 1. (a) The architecture of the proposed D2AN. (b) The structure of the Delay-Doppler attention layer.

1) *Preprocessing*: The sparse complex-valued channel matrix $\mathbf{H}_p \in \mathbb{C}^{N_{f,p} \times N_{t,p}}$ is first converted into a real-valued tensor $\mathbf{D}^{(0)} \in \mathbb{R}^{N_{f,p} \times N_{t,p} \times 2}$ by stacking its real and imaginary parts.

2) *Delay-Doppler Attention Module*: $\mathbf{D}^{(0)}$ is then fed into N_L cascaded attention layers to capture channel correlations.

The proposed attention layer is inspired by the channel autocorrelation functions. Specifically, according to (6) and (7), both temporal and spectral correlations can be viewed as superpositions of infinitely many complex exponential bases weighted by $S(\nu)$ and $P(\tau)$. However, $S(\nu)$ and $P(\tau)$ are typically unknown, and the infinite number of basis components make direct modeling intractable. To address the issue, we sample a set of complex exponential bases and employ FNNs to learn to combine them and generate attention coefficients.

For notational simplicity, the hidden representation at the l -th layer is denoted by $\mathbf{D}^{(l)} \in \mathbb{R}^{N_{f,p} \times N_{t,p} \times C_l}$, where C_l denotes the feature dimension of the layer. For a given resource element (k, n) with $k = 1, \dots, N_{f,p}$ and $n = 1, \dots, N_{t,p}$, the corresponding input feature vector is denoted as $\mathbf{d}_{k,n}^{(l)} \in \mathbb{R}^{C_l}$, which is the entry of $\mathbf{D}^{(l)}$ at position (k, n) . The update procedure of each attention layer is described as follows.

Attention weight generation: Motivated by (6) and (7), we first construct basis vectors by sampling in the delay and Doppler domains. Specifically, the delay and Doppler values are uniformly sampled from the corresponding integration intervals, i.e., $[0, \tau_{\max}]$ and $[-\nu_{\max}, \nu_{\max}]$, respectively. Then, the frequency and time domain basis vectors are defined as

$$\text{rcle}_\tau(k, k') = [e^{j2\pi\tau_1\Delta f(k, k')}, \dots, e^{j2\pi\tau_{M_\tau}\Delta f(k, k')}]^T, \quad (8)$$

$$\mathbf{e}_\nu(n, n') = [e^{-j2\pi\nu_1\Delta t(n, n')}, \dots, e^{-j2\pi\nu_{M_\nu}\Delta t(n, n')}]^T, \quad (9)$$

where M_τ and M_ν are the numbers of samples of τ and ν , respectively, $\Delta f(k, k')$ and $\Delta t(n, n')$ denote the physical frequency and time offsets between two resource elements indexed by (k, n) and (k', n') , respectively, which depend on the subcarrier spacing, OFDM symbol duration and pilot pattern.

The attention weights are then generated by feeding the above basis vectors into FNNs, and the frequency- and time-domain attention weights are obtained as

$$\mathbf{f}_{k,k'}^{(l)} = f_F^{(l)}(\mathbf{e}_\tau(k, k')), \quad \mathbf{t}_{n,n'}^{(l)} = f_T^{(l)}(\mathbf{e}_\nu(n, n')). \quad (10)$$

where $f_F^{(l)}(\cdot)$ and $f_T^{(l)}(\cdot)$ are implemented by FNNs.

Context aggregation: The generated attention weights are then used to aggregate contextual information along the frequency and time dimensions, and the resulting context vectors for (k, n) are given by

$$\text{rcf}_{k,n}^{(l)} = \sum_{k'=0}^{N_{f,p}-1} \mathbf{f}_{k,k'}^{(l)} \odot \mathbf{d}_{k',n}^{(l-1)}, \quad (11)$$

$$\bar{\mathbf{t}}_{k,n}^{(l)} = \sum_{n'=0}^{N_{t,p}-1} \mathbf{t}_{n,n'}^{(l)} \odot \mathbf{d}_{k,n'}^{(l-1)}. \quad (12)$$

As can be seen, the aggregation process incorporates information from the entire frequency and time ranges, thereby effectively capturing long-range dependencies. Moreover, since the attention weights are learned, the network can adaptively focus on the most relevant regions in the time-frequency plane according to the channel characteristics.

Feature update: The aggregated context vectors are then fused with the original hidden representation through an FNN with a residual connection:

$$\mathbf{d}_{k,n}^{(l)} = f_U(\mathbf{d}_{k,n}^{(l-1)}, \bar{\mathbf{f}}_{k,n}^{(l)}, \bar{\mathbf{t}}_{k,n}^{(l)}) + \mathbf{d}_{k,n}^{(l-1)}, \quad (13)$$

where $f_U(\cdot)$ is implemented by an FNN. The residual connection improves training convergence and facilitates the stacking of multiple attention layers.

By stacking N_L attention layers, $\mathbf{D}^{(0)}$ is updated to the final representation $\mathbf{D}^{(N_L)}$.

Unlike conventional attention mechanisms that learn attention weights purely from data [13], the proposed approach explicitly incorporates channel characteristics through predefined basis vectors. As these basis vectors are determined by channel conditions such as user velocity and delay spread, the FNNs can adaptively generate attention weights according to varying channel statistics. This design enables the network to capture underlying correlations and maintain robust performance in dynamic propagation environments. Moreover, D2AN achieves a global receptive field with only two attention layers, while CNNs require deeper architectures to attain a global receptive field, especially when $N_{f,p}$ and $N_{t,p}$ are large.

3) *Upsampling Module*: The upsampling module maps $\mathbf{D}^{(N_L)} \in \mathbb{R}^{N_{f,p} \times N_{t,p} \times C_{N_L}}$ to $\hat{\mathbf{D}} \in \mathbb{R}^{N_f \times N_t \times 2}$ with the target resolution on the full time-frequency grid. Specifically, it contains a transposed convolutional layer for resolution expansion, followed by a convolutional layer with a 1×1 kernel to refine the output features. The kernel size and stride of the transposed convolution are set as $(\frac{N_f}{N_{f,p}}, \frac{N_t}{N_{t,p}})$, such that the output matches the dimensions of the full channel matrix.

4) *Postprocessing*: In this stage, the real-valued network output $\hat{\mathbf{D}} \in \mathbb{R}^{N_f \times N_t \times 2}$ is converted to the complex channel estimate $\hat{\mathbf{H}} \in \mathbb{C}^{N_f \times N_t}$.

Remark 1: For MIMO systems, the Delay-Doppler attention mechanism can still be applied in the time-frequency domain, while additional attention mechanisms can be introduced in the antenna dimension to capture spatial correlations.

C. Model Training

The proposed D2AN is trained in a supervised manner.

Denote the training set as $\mathcal{T} = \{\mathbf{H}^{(n)}\}_{n=1}^{N_{tr}}$, where N_{tr} is the number of training samples. During training, for each sample $\mathbf{H}^{(n)}$, a noisy sparse observation $\mathbf{Y}_p^{(n)}$ is generated according

TABLE I

PARAMETERS FOR SYSTEM SETTING	
Parameter	Value
(N_f, N_t)	(120, 14)
Carrier Frequency	3.5 GHz
Subcarrier Spacing	15 KHz
Pilot Symbols	The 3rd and 12th OFDM symbols [10]
Pilot Spacing	4 subcarriers
Channel Model	CDL-D
Delay Spreads	{10, 100, 300, 1000} ns
User Velocities	{30, 60, 90, 120, 300} km/h
SNR	{0, 5, 10, 15, 20} dB

TABLE II

HYPER-PARAMETERS	
Parameter	Value
N_L, C_l	(4, 16)
(M_f, M_t)	(128, 32)
$f_F^{(l)}(\cdot), f_T^{(l)}(\cdot)$	Two-layer FNN with C_l neurons
Transposed Conv. Kernel	16 kernels (4×7)
Conv. Kernel	2 kernels (1×1)
Epoch	200
Batch Size	256
Learning Rate	1×10^{-3}

to (2), with the noise power randomly drawn from a predefined set. $\mathbf{Y}_p^{(n)}$ is then fed into the network to produce the estimate $\hat{\mathbf{H}}^{(n)}$. The loss function during training is

$$\mathcal{L} = \frac{1}{N_{tr}} \sum_{i=1}^{N_{tr}} \|\mathbf{H}^{(i)} - \hat{\mathbf{H}}^{(i)}\|_F^2. \quad (14)$$

IV. SIMULATION RESULTS

In this section, we evaluate the effectiveness and robustness of the proposed D2AN via simulations.

A. Simulation Setup

We consider a SISO-OFDM system with parameters listed in Table I, where the pilot overhead is $\frac{2}{14} \times \frac{1}{4} \approx 3.57\%$. To ensure the model generalizes across diverse environments, the channel realizations for training are generated with varying delay spreads, user velocities and signal-to-noise ratios (SNRs). The training and test datasets contain 40,000 and 10,000 samples, respectively.

The hyper-parameters of the D2AN are listed in Table II.

The proposed D2AN is compared with several learning-based methods, including ChannelNet [4], SisRafNet [7], ReEsNet [5], and AdaFortiTran [10]. In addition, we also compare with conventional LS, LMMSE, and ALMMSE estimators that perform sequential one-dimensional LMMSE in the frequency and time domains. For each scenario, the performance is evaluated and averaged on 2,000 samples.

B. Channel Estimation Error

Fig. 2 illustrates the MSE of the compared methods under different SNR levels. All learning-based methods outperform conventional approaches, and the proposed D2AN achieves the lowest MSE by effectively exploiting channel statistics and capturing channel correlations. To quantify the contribution of each component, we compare D2AN with its two variants: (a) replacing the proposed attention mechanism with the dot-product attention in [13] (denoted as “Proposed with D.P.A.”), and (b) removing the residual connections (denoted

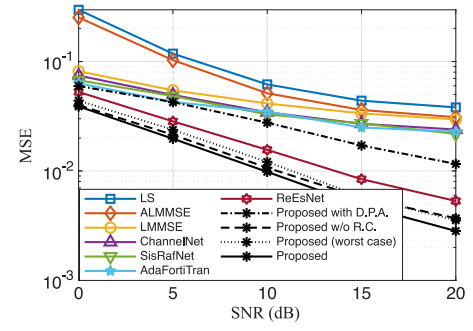


Fig. 2. MSE performance versus SNR.

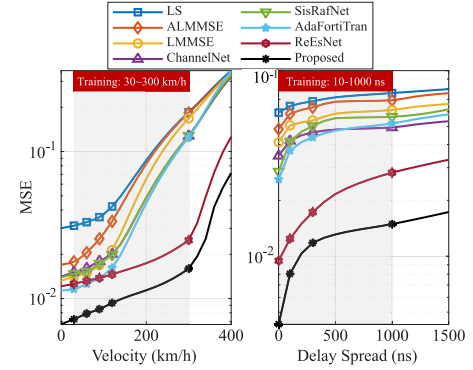


Fig. 3. MSE performance versus delay spread or velocity.

as “Proposed w/o R.C.”). The results indicate that incorporating domain knowledge contributes more significantly to performance improvement than the residual connections.

In Fig. 2, we also consider the case where channel-specific delay spreads and velocities are unavailable, and fixed global worst-case parameters (i.e., 1000 ns and 300 km/h) are fed into the D2AN (denoted as “Proposed (worst case)”). As can be seen, the proposed method still outperforms the baselines, indicating that the performance gain mainly stems from the proposed attention mechanism incorporating prior knowledge, rather than from the availability of side information.

The robustness of D2AN under varying channel conditions is illustrated in Fig. 3, which evaluates the estimation accuracy with respect to delay spread and user velocity, respectively. As the delay spread or user velocity increases, the performance of all methods degrades due to reduced spectral or temporal channel correlation. Nevertheless, D2AN consistently achieves the lowest MSE among all methods, even though it is trained on discrete delay spreads (as indicated by the markers in Fig. 3) and velocities but tested on continuous values, including those not observed during training and even beyond the training range. This indicates that the D2AN can adapt to varying channel statistics without frequent retraining, enabling reliable real-time estimation under dynamic channel conditions.

In Fig. 4, we evaluate the MSE of D2AN across subcarriers when the SNR is 10 dB. As can be seen, the MSE is higher near the band edges and lower in the middle subcarriers. This is because the middle subcarriers benefit from channel correlation on both sides, while the edge subcarriers can only exploit one-sided correlation information, making estimation slightly less accurate.

Table III compares the MSE of the D2AN with the best learning-based and non-learning methods, i.e., ReEsNet and

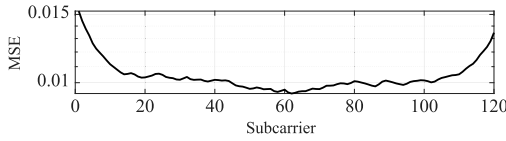


Fig. 4. MSE performance of D2AN across subcarriers.

TABLE III
MSE PERFORMANCE VERSUS PILOT SPACING ($\times 10^{-3}$)

Pilot Spacing (subcarriers)	2	4	8	12
LMMSE	35.8	41.3	50.4	55.5
ReEsNet	13.1	17.2	23.7	28.6
Proposed	6.6	10.7	17.0	22.0

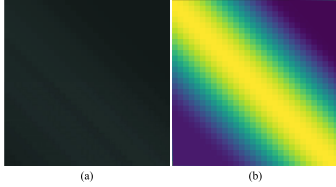


Fig. 5. Heatmaps of the learned attention matrices for delay spreads of (a) 10 ns and (b) 300 ns.

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY

Method	Parameters (M)	Inference Time (ms)
LS	\	0.0459
LMMSE	\	0.119
ALMMSE	\	0.0529
ChannelNet	0.644	0.509
SisRafNet	0.589	1.537
ReEsNet	0.0273	0.208
AdaFortiTran	0.235	0.885
D2AN (Proposed)	0.0276	0.860

LMMSE, under different pilot spacings. As the pilot spacing increases, the performance of all methods degrades, while the D2AN consistently achieves the best performance.

C. Interpretability

To interpret the adaptability of the proposed method, we consider the frequency domain and construct an attention matrix $\mathbf{F} = [\text{mean}(\mathbf{f}_{k,k'}^{(1)})]_{k,k'}$, where each element represents the average attention weight after training from the k' -th to the k -th subcarrier in the first layer. The corresponding heatmaps under different delay spreads are shown in Fig. 5. When the delay spread is 10 ns, the channel exhibits strong frequency correlation, and the attention matrix is nearly uniform, indicating that all subcarriers contribute similarly. In contrast, at 300 ns, the diagonal elements become dominant, implying that the network concentrates more on neighboring subcarriers, which is consistent with the underlying channel correlation structure.

D. Computational Complexity

In this subsection, we evaluate the computational efficiency of the proposed D2AN in terms of the number of trainable parameters and inference time. The evaluations are conducted on a computer with an AMD Ryzen 7 9800 $\times$ 3D CPU and an NVIDIA GeForce RTX 5080 GPU. The results are summarized in Table IV.

As can be seen, the D2AN and ReEsNet employ a significantly smaller number of parameters than the other deep

learning-based methods, since they perform upsampling at the final stage of the model [5].

In terms of inference time, D2AN achieves the lowest inference time of 0.86 ms among the DNNs with global receptive fields, namely SisRafNet and AdaFortiTran. Although CNN-based methods such as ChannelNet and ReEsNet incur a shorter inference time, they are less effective to capture long-range dependencies than D2AN as analyzed in Section III-B.

V. CONCLUSION AND FUTURE WORK

In this letter, we proposed the Delay-Doppler Attention Network (D2AN) for channel estimation, with an architecture specifically designed to capture the intrinsic delay and Doppler correlations of wireless channels. A specialized attention mechanism was developed by leveraging physical basis functions sampled from the delay-Doppler domain, thereby embedding the structure of channel correlation functions into the network and enabling adaptive adjustment of attention coefficients under varying propagation environments. Simulation results validate the effectiveness of the proposed D2AN, demonstrating superior estimation accuracy and enhanced robustness compared with existing state-of-the-art methods.

Future work will investigate the extension of the proposed method to MIMO systems and the enhancement of its generalizability across diverse channel models.

REFERENCES

- [1] L. Cimini, "Analysis and simulation of a digital mobile channel using orthogonal frequency division multiplexing," *IEEE Trans. Commun.*, vol. C-33, no. 7, pp. 665–675, Jul. 1985.
- [2] J.-C. Lin, "Least-squares channel estimation for mobile OFDM communication on time-varying frequency-selective fading channels," *IEEE Trans. Veh. Technol.*, vol. 57, no. 6, pp. 3538–3550, Nov. 2008.
- [3] V. Savaux and Y. Louët, "LMMSE channel estimation in OFDM context: A review," *IET Signal Process.*, vol. 11, no. 2, pp. 123–134, Apr. 2017.
- [4] M. Soltani, V. Pourahmadi, A. Mirzaei, and H. Sheikhzadeh, "Deep learning-based channel estimation," *IEEE Commun. Lett.*, vol. 23, no. 4, pp. 652–655, Apr. 2019.
- [5] L. Li, H. Chen, H.-H. Chang, and L. Liu, "Deep residual learning meets OFDM channel estimation," *IEEE Wireless Commun. Lett.*, vol. 9, no. 5, pp. 615–618, May 2020.
- [6] Y. Liao, Y. Hua, X. Dai, H. Yao, and X. Yang, "ChanEstNet: A deep learning based channel estimation for high-speed scenarios," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.
- [7] A. S. M. M. Jameel, A. Malhotra, A. E. Gamal, and S. Hamidi-Rad, "Deep OFDM channel estimation: Capturing frequency recurrence," *IEEE Commun. Lett.*, vol. 28, no. 3, pp. 562–566, Mar. 2024.
- [8] W. Gao, W. Zhang, L. Liu, and M. Yang, "Deep residual learning with attention mechanism for OFDM channel estimation," *IEEE Wireless Commun. Lett.*, vol. 14, no. 2, pp. 250–254, Feb. 2025.
- [9] M. Gok, J. Zhou, S. Bhattacharjee, S. Sambhwani, and H. Niu, "CHAST: Attention aided SISO OFDM channel estimation," in *Proc. NeurIPS*, 2025. [Online]. Available: <https://openreview.net/pdf?id=Ei7OTmEQNi>
- [10] B. Guler and H. Jafarkhani, "AdaFortiTran: An adaptive transformer model for robust OFDM channel estimation," in *Proc. ICC - IEEE Int. Conf. Commun.*, Jun. 2025, pp. 3797–3802.
- [11] F. Liu, P. Jiang, J. Zhang, W. Wang, C.-K. Wen, and S. Jin, "PD-CEViT: A novel pilot pattern design and channel estimation network for OFDM systems," *IEEE Trans. Commun.*, vol. 73, no. 6, pp. 4363–4377, Jun. 2025.
- [12] N. Shlezinger, J. Whang, Y. C. Eldar, and A. G. Dimakis, "Model-based deep learning," *Proc. IEEE*, vol. 111, no. 5, pp. 465–499, May 2023.
- [13] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, vol. 30, 2017. [Online]. Available: <https://dl.acm.org/doi/10.5555/3295222.3295349>